UID EX 10: Create a data visualization (e.g., pie charts, bar graphs) for the most watched movie genres in Tamil Nadu using JavaScript

AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for analyzing the most watched movie genres in Tamil Nadu using JavaScript. JavaScript chart libraries such as Chart.js are commonly used to render pie and bar charts inside HTML canvas elements.

PROCEDURE:

Step 1:

Set Up Your HTML File

Create an HTML file that contains the basic structure of the webpage. Add heading text for the title and include two anvas> elements, one for the pie chart and one for the bar chart. Then include the Chart.js CDN link and connect your external JavaScript file. Chart.js charts are typically rendered inside HTML anvas> elements.

index.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0
  <title>Most Watched Movies in Tamil Nadu</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <h1>Most Watched Movies in Tamil Nadu</h1>
    <p class="subtitle">Data Visualization using JavaScript and Chart.

    <div class="chart-container">
      <div class="chart-box">
        <h2>Genre Popularity Distribution</h2>
        <canvas id="pieChart"></canvas>
      </div>

      <div class="chart-box">
        <h2>Most Watched Movie Genres</h2>
        <canvas id="barChart"></canvas>
      </div>
    </div>
  </div>

  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <script src="script.js"></script>
</body>
</html>
```

Step 2:

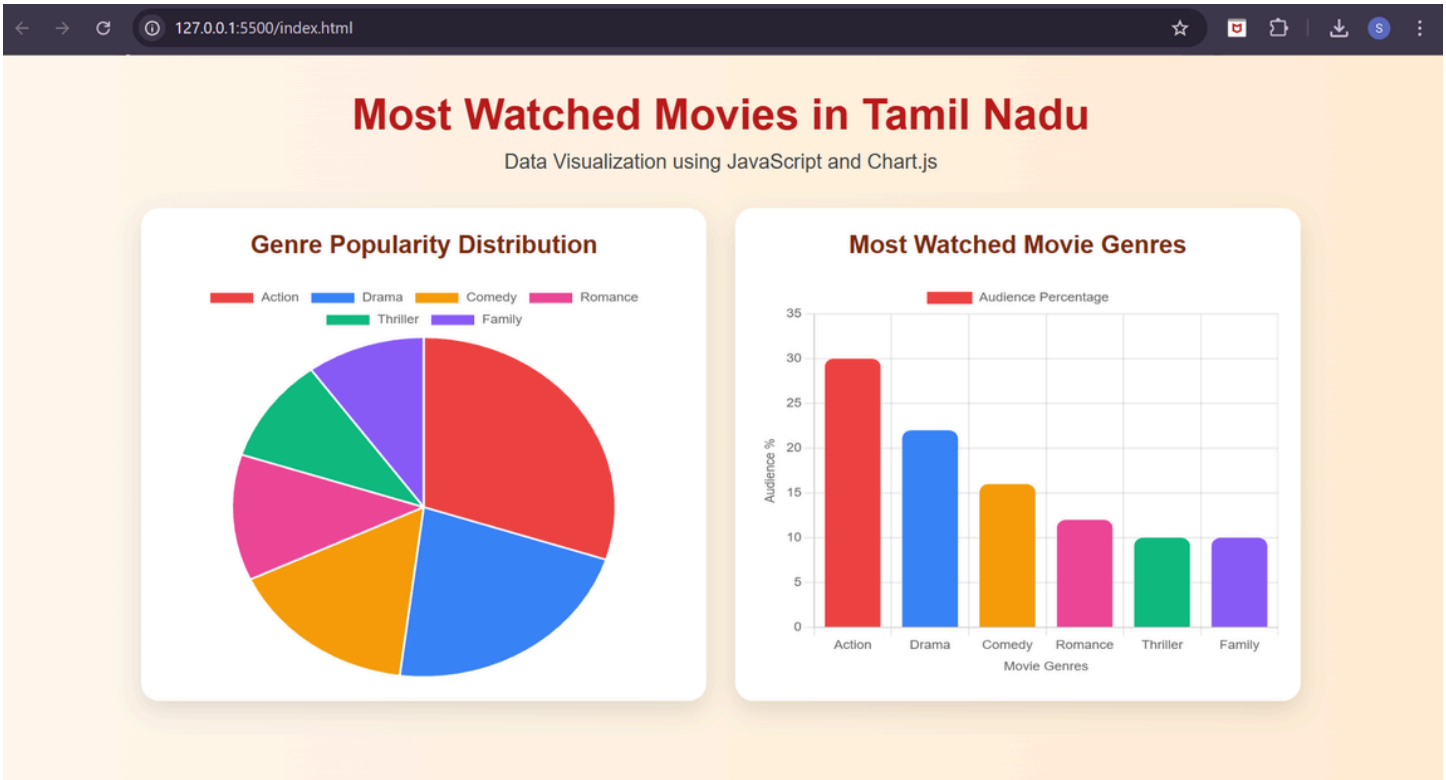
Create the JavaScript File for Charts

Next, create a JavaScript file (script.js) to handle the data visualization logic. Define the movie-related data, such as genres and audience percentages, then use Chart.js to create a pie chart for genre distribution and a bar chart for comparing popularity across genres. Chart.js charts are created by instantiating a Chart object with chart type, data, and options.

```js
const movieData = {
  labels: ["Action", "Drama", "Comedy", "Romance", "Thriller", "F
  datasets: [
    {
      label: "Most Watched",
      data: [30, 22, 16, 12, 10, 10],
      backgroundColor: [
        "#ef4444",
        "#3b82f6",
        "#f59e0b",
        "#ec4899",
        "#10b981",
        "#8b5cf6"
      ],
      borderColor: "#ffffff",
      borderWidth: 2
    }
  ]
};

const pieCtx = document.getElementById("pieChart").getContext("2d");
new Chart(pieCtx, {
  type: "pie",
  data: movieData,
  options: {
    responsive: true,
    maintainAspectRatio: false,
    plugins: {
      legend: {
        position: "top"
      },
      tooltip: {
        callbacks: {
          label: function(context) {
            return context.label + ": " + context.raw + "%";
```

OUTPUT:



- A **pie chart** showing the distribution of the most watched movie genres in Tamil Nadu. Pie charts are used to represent proportional data visually.
- A **bar graph** showing comparison of audience percentage across movie genres such as Action, Drama, Comedy, Romance, Thriller, and Family. Bar charts are commonly used for

category comparison.

- The webpage displays both charts clearly with proper titles, colors, and responsive layout using HTML, CSS, and JavaScript. Responsive chart layouts are a common use case supported by Chart.js